

# Tutorial 8

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## 1 Parity Revisited

Consider a particle that is localised around the position  $\vec{r}$ . Parity operator is defined through it's action on position eigenket.

$$\hat{\Pi} |\vec{r}\rangle = |-\vec{r}\rangle \quad (1)$$

i.e parity operator inverts the location of the particle.

### Properties of Parity operator

- (a) Parity operator is a unitary operator as well as a hermitian operator.
- (b) Parity operator is the inverse of its own.
- (c) Eigenvalues of the parity operator are  $\pm 1$ .

### Wavefunction under parity

- (i) Show that the wavefunction of a system described by the vector  $|\psi\rangle$  transforms as:

$$\langle x | \hat{\Pi} |\psi\rangle = \psi(-x) \quad (2)$$

Using  $|\psi\rangle \equiv |p\rangle$ , Show that:

$$\hat{\Pi} |p\rangle = |-p\rangle \quad (3)$$

- (ii) Show that the wave functions corresponding to the eigenvector of Parity operator are even/odd functions.

### Commutator and anti commutator with parity operator

- (a) Under Passive mode of transformation, it is assumed that state ket remains the same whereas the operators transform. Show that position and momentum operator transform as:

$$\hat{\Pi}^\dagger \hat{x} \hat{\Pi} = -\hat{x} \quad \hat{\Pi}^\dagger \hat{p} \hat{\Pi} = -\hat{p} \quad (4)$$

- (b) Using the above relation, show that position and momentum operator anti commutes with parity operator.
- (c) Show that angular momentum operator commutes with the parity operator

**Aside** A Hamiltonian describing a system is said to be parity invariant (or possess parity symmetry) if:

$$[\hat{\Pi}, \hat{\mathcal{H}}] = 0 \quad (5)$$

## 2 Symmetry

A particle is in a potential

$$V(x) = V_0 \sin(2\pi x/a) \tag{6}$$

which is invariant under the translations  $x \rightarrow x + ma$ , where  $m$  is an integer. Is momentum conserved? Why not?